Oxidative Deamination

Oxidative Deamination is a process in which the alpha amino group of the amino acid is removed as ammonia, coupled with oxidation. Most of the amino acids dump their amino groups to alpha ketoglutarate and pyruvate forming glutamate and alanine, respectively.

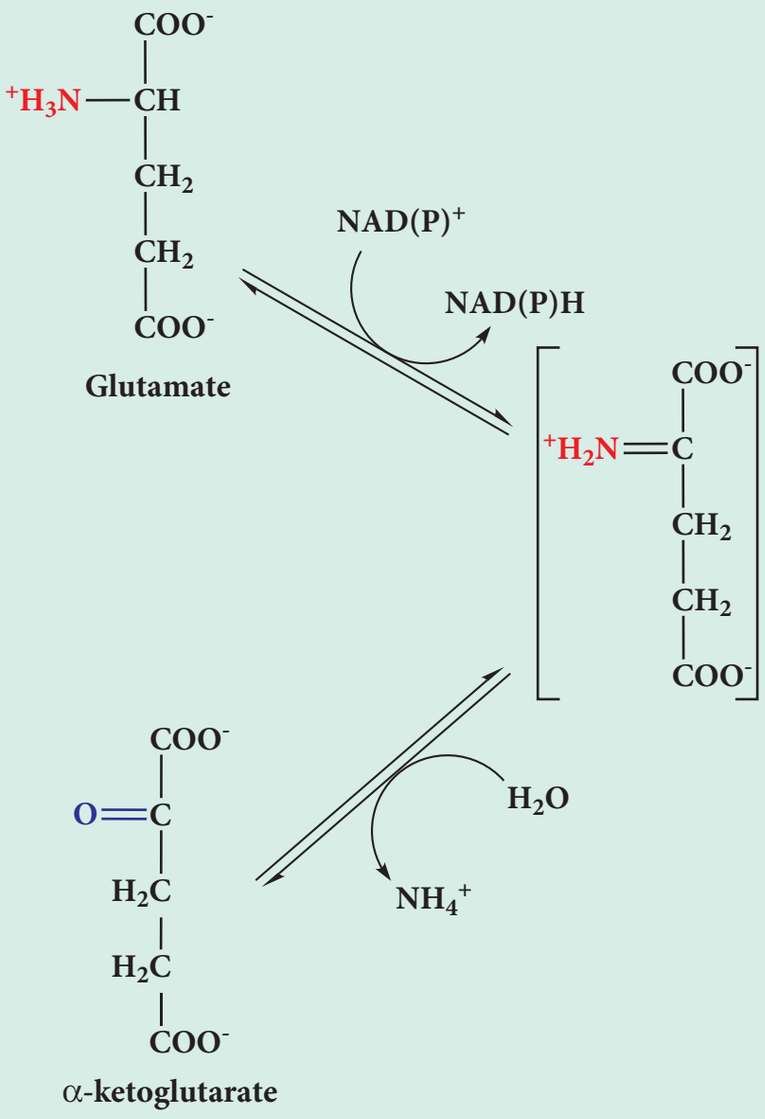

Figure 4.6. Oxidative deamination of glutamate

Oxidative deamination involving the removal of amino group of glutamate as ammonia is the key reaction involved in delivery of amino groups to the liver for urea synthesis. The reaction is catalysed by glutamate dehydrogenase (GDH) with NAD^+/ $NADP^+$ as the co-enzyme and this reaction is freely reversible. The enzyme is present in liver and kidneys and it plays a central role in nitrogen metabolism. Liver GDH activity is allosterically inhibited by ATP and GTP (Fig 4.6).

As glutamate acts as a collection centre for all alpha amino groups by transamination and subsequently glutamate liberates ammonia, the process is collectively called as trans-deamination.

L-amino acid oxidases and D-amino acid oxidases are also involved in oxidative deamination.

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